Micro Set 6. PC vs Monopoly — Model Answer Scheme

Question 1 [2 marks]

Two characteristics from Extract 1.1:

- **Large number of small firms:** approximately 6000 stalls operating, each with less than 0.1% market share.
- **Low barriers to entry/exit:** entry costs are S$30000–80000 in equipment; "NEA provides a relatively transparent and competitive tender process."

[1 mark per feature.]

Examiner Note. Other valid features: homogeneous-within-cuisine product (price-taking), low search costs (perfect information approximation), free exit (failing stalls close). Stating "many sellers" alone is too vague.

Question 2 [2 marks]

A **natural monopoly** exists when one firm can supply the entire market at lower average cost than two or more firms could, due to large economies of scale and high fixed network costs. *[1 mark.]*

Evidence: (i) "the fixed costs of duplicating this infrastructure are prohibitive — over S$10 billion" (high fixed costs); (ii) "average cost falls continuously over the relevant range of output" (sustained EOS). *[1 mark for both features.]*

Question 3 [4 marks]

Why hawker profit margins compete down to 5–12%.

The hawker sector approximates perfect competition. Free entry means that if any stall earns supernormal profit, new entrants are attracted into the market. *[1 mark for free-entry mechanism.]*

New entry expands market supply, putting downward pressure on prices and reducing the abnormal profits of incumbents. Entry continues until profits are competed down to normal profit (just covering opportunity cost). *[1 mark for adjustment to normal profit.]*

The empirical 5–12% margin reflects normal profit: it covers the opportunity cost of the stall operator's time, capital, and risk-bearing, but does not include rents from monopoly power. Variation within this range reflects location quality and cuisine differentiation rather than market power. *[1 mark for interpretation.]*

Extract 1.1 also notes the disciplining effect of low search costs: "If a stall consistently overprices or underdelivers, customers shift to a neighbouring stall within hours." This rapid demand reallocation reinforces price discipline in the short run. *[1 mark for evidence anchor.]*

Examiner Note. The full answer combines the long-run entry mechanism with the short-run demand-side discipline. Either alone earns 2–3 marks.

Question 4 [4 marks]

EMA regulation at 6% return on capital.

An unregulated monopoly sets price where MR = MC, generating supernormal profit and producing a quantity $Q_m < Q^*$ where $P_m > MC$ (allocative inefficiency). Consumer surplus is reduced and welfare is lost. *[1 mark for unregulated monopoly outcome.]*

Under the EMA framework, SP Group's allowed return is capped at 6% on capital. This limits the producer surplus extracted to the cost of capital plus normal return, preventing supernormal-profit extraction at consumers' expense. *[2 marks for return cap.]*

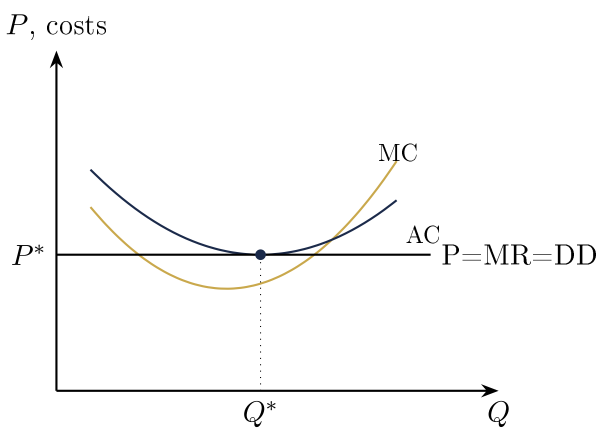
The regulated outcome combines EOS-driven low average cost (the monopoly's natural advantage) with consumer-welfare protection (the regulator's discipline). Compared to unregulated monopoly, output is closer to the socially efficient level and price is closer to MC; compared to no monopoly at all (hypothetical duplication), unit costs are lower. *[1 mark for combined welfare outcome.]*

Examiner Note. Top answers frame regulation as solving a *joint* problem: capture EOS benefits while preventing rent extraction. Without this framing the answer reads as merely "capping profit."

Question 5 [6 marks]

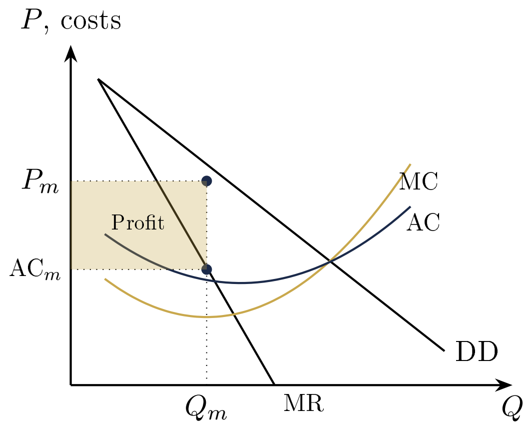
PC vs Monopoly: long-run equilibrium.

Perfect competition (hawker stalls):



In long-run equilibrium, the perfectly competitive firm produces where $P = MR = MC = \min AC$. Profit is normal (zero supernormal profit). Free entry ensures that any short-run supernormal profit attracts new firms, shifting industry supply until prices fall to the minimum-AC level. The firm is both allocatively efficient ($P = MC$) and productively efficient (operating at $\min AC$). *[2 marks for diagram + interpretation.]*

Monopoly (SP Group, unregulated):



The monopoly maximises profit where $MR = MC$, producing Q_m at price $P_m > MC$. With barriers to entry, supernormal profit (shaded rectangle) persists in the long run. Output is below the allocative-efficient level (where $P = MC$), and the firm need not operate at minimum AC. *[2 marks.]*

Contrast.

- Price: PC = MC; Monopoly $P > MC$.
- Output: PC at allocative-efficient level; Monopoly produces less, creating DWL.
- Profit: PC earns normal profit; Monopoly earns supernormal profit.
- Efficiency: PC is both allocatively and productively efficient; Monopoly is allocatively inefficient.

[2 marks for explicit comparison.]

Examiner Note. Both diagrams must show all relevant cost curves (MC, AC) and demand structures (horizontal P=MR for PC; downward-sloping DD with MR below for monopoly). The shaded profit rectangle is essential for the monopoly diagram.

Question 6 [6 marks]

Why competition is unsuitable for electricity distribution.

Reason 1: Wasteful duplication and EOS loss. The fixed cost of building a competing electricity grid is over S$10 billion (Extract 1.1). Allowing two firms would require either duplicate infrastructure (each company building parallel cables, substations, transformers) or shared use with complex pricing. Either way, total fixed cost is far higher than a single network, raising unit costs and consumer prices. The economies of scale in transmission are so large that "average cost falls continuously over the relevant range," meaning a single firm is unambiguously more cost-efficient than two. *[2 marks.]*

Reason 2: Coordination and reliability failure. Electricity distribution requires coordinated voltage management, fault response, and load balancing across the entire network. Two competing operators would face significant coordination costs and reliability risks — a failure on one network's section could cascade to the other without unified control. The EMA explicitly observes that "allowing free-market competition in electricity distribution would simply not work" (Extract 1.1). *[2 marks.]*

Reason 3: Investment irreversibility and crowding out. Network investments are highly specific (sunk cost) and have decades-long payback periods. Competing investments would require committing capital that, if the rival firm prevails, becomes worthless. This deters efficient investment ex ante. A single regulated monopoly avoids this duplication risk and channels investment efficiently. *[2 marks.]*

> **Examiner Note.** Strong answers connect natural-monopoly theory (EOS) with practical coordination realities (grid management). Listing only "it's expensive" caps at 3 marks.

End of Set 6 — Model Answers